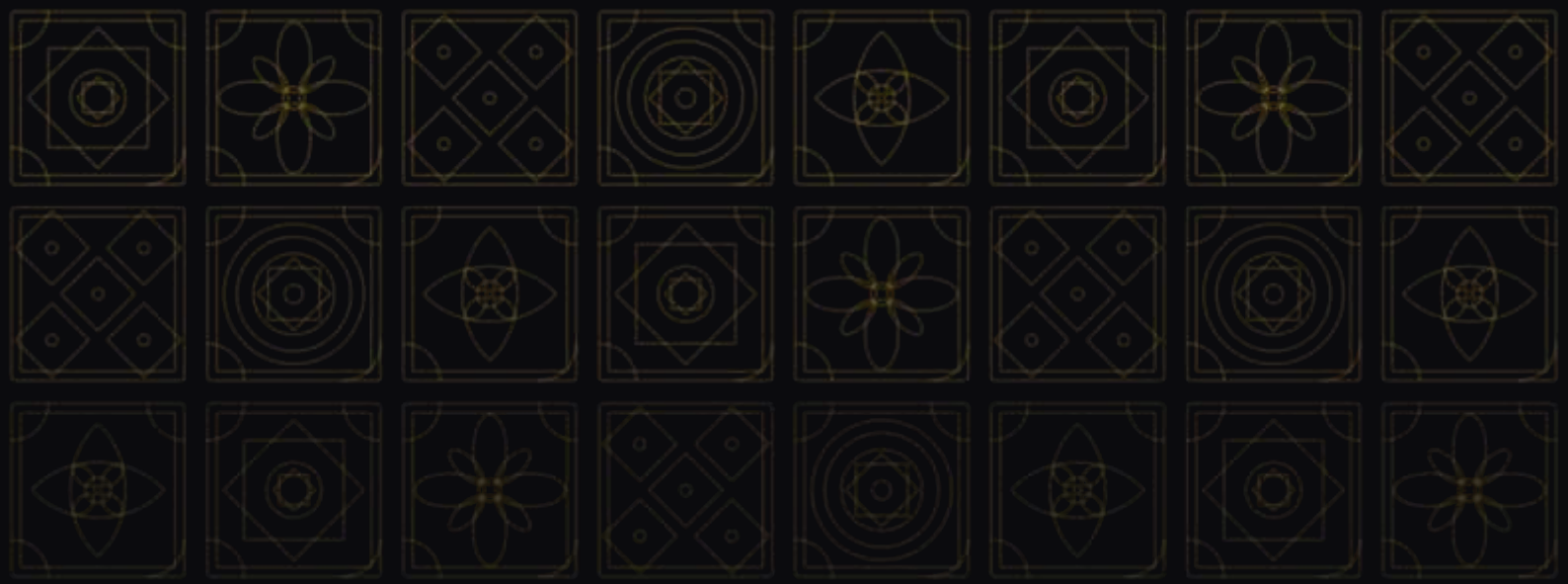




TRAINING PROGRAMME

# Classical AI

Search, logic and probabilistic reasoning: the foundations of artificial intelligence

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**90** HOURS

**FOR** University students in computer science and engineering



**Mohamed Aziz BEN HAH**

[mohamedazizbenhaha.netlify.app](http://mohamedazizbenhaha.netlify.app)

## Overview

This one-semester academic course covers the foundations of artificial intelligence: rational agents, search and problem solving, game playing, knowledge representation and logic, and reasoning under uncertainty and over time. It follows the structure of Russell and Norvig's *Artificial Intelligence: A Modern Approach* (4th edition). Each part combines lectures that derive the algorithms with an implementation mini-project in Python, so that students both understand and build each method. The course closes with a capstone project: an AI system combining search, logic or probabilistic reasoning, proposed, developed and presented by each team.

## At a glance

| #  | PHASE                             | FOCUS                                                                      | HOURS |
|----|-----------------------------------|----------------------------------------------------------------------------|-------|
| 01 | Introduction & Intelligent Agents | What AI is, agents and environments, rationality                           | 9 h   |
| 02 | Search & Problem Solving          | Uninformed and heuristic search, local search, evolutionary methods, CSPs  | 18 h  |
| 03 | Game Playing                      | Minimax, alpha-beta pruning, evaluation functions, Monte Carlo tree search | 6 h   |
| 04 | Knowledge Representation & Logic  | Propositional and first-order logic, inference, knowledge engineering      | 15 h  |
| 05 | Probabilistic Reasoning           | Probability, Bayesian networks, exact and approximate inference            | 15 h  |
| 06 | Reasoning over Time               | Markov processes, hidden Markov models, filtering, smoothing, Viterbi      | 9 h   |
| 07 | Final Project                     | A capstone AI system, proposed, developed and presented                    | 9 h   |
|    | Buffer                            | Catch-up, revision, extra lab time                                         | 9 h   |
|    | Total                             |                                                                            | 90 h  |

## Key facts

|                 |                                                                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUDIENCE        | University students in computer science and engineering                                                                                                                 |
| PREREQUISITES   | Python programming (course: Python Programming Fundamentals), or equivalent; discrete mathematics and basic probability                                                 |
| DURATION        | 90 h: 81 h of chapters + 9 h buffer, over one academic semester                                                                                                         |
| FORMAT          | Lectures that derive each algorithm, followed by implementation mini-projects in Python                                                                                 |
| LAB ENVIRONMENT | One laptop per student with Python 3, Jupyter and VS Code; course code and datasets in a shared repository; no account or cost                                          |
| ASSESSMENT      | Six mini-projects and a capstone project presented to the class; certificate of completion for students who complete both; grading follows the host institution's rules |
| OUTCOME         | Students can formulate AI problems, implement and compare search, logic and probabilistic algorithms, and justify which method fits a problem                           |

## PHASE 01

9 H

# Introduction & Intelligent Agents

Students see what AI is, where it came from and where it is used, including its ethical questions, then learn the agent framework used throughout the course: environments, rationality and agent architectures. Mini-project: a simple agent designed and simulated in a grid environment.

| #   | CHAPTER              | CONTENT                                                | FORMAT             | HOURS |
|-----|----------------------|--------------------------------------------------------|--------------------|-------|
| 1.1 | What AI Is           | Definitions, history, applications, ethics             | Lecture + Workshop | 3 h   |
| 1.2 | Intelligent Agents   | Agents, environments, rationality, agent architectures | Lecture + Lab      | 3 h   |
| 1.3 | Mini-Project: Agents | Design and simulate a simple agent                     | Project            | 3 h   |

## PHASE 02

18 H

# Search & Problem Solving

Students formulate problems as state spaces and solve them with uninformed and heuristic search, local search, genetic algorithms and particle swarms, and constraint satisfaction. Mini-project: a planning or optimisation problem solved and compared with at least two search methods.

| #   | CHAPTER                      | CONTENT                                                                 | FORMAT        | HOURS |
|-----|------------------------------|-------------------------------------------------------------------------|---------------|-------|
| 2.1 | Uninformed Search            | Problem formulation, BFS, DFS, uniform-cost search, iterative deepening | Lecture + Lab | 3 h   |
| 2.2 | Heuristic Search             | Greedy best-first, A*, admissibility and consistency                    | Lecture + Lab | 3 h   |
| 2.3 | Local Search                 | Hill climbing, simulated annealing, local beam search                   | Lecture + Lab | 3 h   |
| 2.4 | Evolutionary & Swarm Methods | Genetic algorithms, particle swarm optimisation                         | Lecture + Lab | 3 h   |
| 2.5 | Constraint Satisfaction      | CSPs, backtracking, arc consistency                                     | Lecture + Lab | 3 h   |
| 2.6 | Mini-Project: Search         | Solve a planning or optimisation problem with search                    | Project       | 3 h   |

## MATHS COVERED IN THIS PHASE

Graph theory (state spaces, paths) · Complexity (time and space of search algorithms) · Optimisation (objective functions, local optima, annealing schedules)

## PHASE 03

6 H

## Game Playing

Students search against an opponent with minimax and alpha-beta pruning, design evaluation functions, and use Monte Carlo tree search when the game tree is too large. Mini-project: a game-playing agent evaluated against baseline players.

| #   | CHAPTER                  | CONTENT                                                                    | FORMAT        | HOURS |
|-----|--------------------------|----------------------------------------------------------------------------|---------------|-------|
| 3.1 | Adversarial Search       | Minimax, alpha-beta pruning, evaluation functions, Monte Carlo tree search | Lecture + Lab | 3 h   |
| 3.2 | Mini-Project: Game Agent | Build and evaluate a game-playing agent                                    | Project       | 3 h   |

**MATHS COVERED IN THIS PHASE**

Game theory (zero-sum games, minimax value) · Probability (Monte Carlo estimates, UCB1 in MCTS)

## PHASE 04

15 H

## Knowledge Representation & Logic

Students represent knowledge in propositional and first-order logic and derive new facts with resolution, DPLL, forward and backward chaining and unification. Mini-project: a small knowledge base and reasoner that answers queries in a chosen domain.

| #   | CHAPTER                                    | CONTENT                                         | FORMAT        | HOURS      |
|-----|--------------------------------------------|-------------------------------------------------|---------------|------------|
| 4.1 | <b>Propositional Logic</b>                 | Syntax, semantics, entailment                   | Lecture + Lab | <b>3 h</b> |
| 4.2 | <b>Propositional Inference</b>             | Resolution, DPLL, forward and backward chaining | Lecture + Lab | <b>3 h</b> |
| 4.3 | <b>First-Order Logic</b>                   | Syntax, semantics, knowledge engineering        | Lecture + Lab | <b>3 h</b> |
| 4.4 | <b>First-Order Inference</b>               | Unification, chaining, resolution               | Lecture + Lab | <b>3 h</b> |
| 4.5 | <b>Mini-Project: Knowledge-Based Agent</b> | Build a small knowledge base and reasoner       | Project       | <b>3 h</b> |

**MATHS COVERED IN THIS PHASE**

Formal logic (syntax, semantics, entailment, soundness and completeness) · Satisfiability · Unification

PHASE 05

15 H

# Probabilistic Reasoning

Students reason under uncertainty with probability and Bayes' rule, model dependencies with Bayesian networks, and answer queries with exact inference and with sampling methods. Mini-project: a Bayesian network for a real decision problem, queried and checked against data.

| #   | CHAPTER                               | CONTENT                                        | FORMAT        | HOURS      |
|-----|---------------------------------------|------------------------------------------------|---------------|------------|
| 5.1 | <b>Uncertainty &amp; Probability</b>  | Probability axioms, Bayes' rule, independence  | Lecture + Lab | <b>3 h</b> |
| 5.2 | <b>Bayesian Networks</b>              | Structure, conditional independence, semantics | Lecture + Lab | <b>3 h</b> |
| 5.3 | <b>Exact Inference</b>                | Enumeration, variable elimination              | Lecture + Lab | <b>3 h</b> |
| 5.4 | <b>Approximate Inference</b>          | Sampling, likelihood weighting, MCMC           | Lecture + Lab | <b>3 h</b> |
| 5.5 | <b>Mini-Project: Bayesian Network</b> | Model and query a real decision problem        | Project       | <b>3 h</b> |

## MATHS COVERED IN THIS PHASE

Probability (axioms, conditional probability, Bayes' rule, independence) · Sampling and Markov chain Monte Carlo

## PHASE 06

9 H

## Reasoning over Time

Students model processes that change over time with Markov chains and hidden Markov models, and infer hidden states with filtering, smoothing and the Viterbi algorithm. Mini-project: sequence labelling or tracking with a hidden Markov model.

| #   | CHAPTER                           | CONTENT                                    | FORMAT        | HOURS      |
|-----|-----------------------------------|--------------------------------------------|---------------|------------|
| 6.1 | <b>Temporal Models &amp; HMMs</b> | Markov processes, hidden Markov models     | Lecture + Lab | <b>3 h</b> |
| 6.2 | <b>Inference over Time</b>        | Filtering, smoothing, Viterbi              | Lecture + Lab | <b>3 h</b> |
| 6.3 | <b>Mini-Project: HMM</b>          | Sequence labelling or tracking with an HMM | Project       | <b>3 h</b> |

**MATHS COVERED IN THIS PHASE**

Markov chains · Dynamic programming (forward-backward, Viterbi)

## PHASE 07

9 H

# Final Project

Teams of two or three propose and build an AI system that combines at least two parts of the course, reusing their mini-projects, then present it to the class.

| #   | CHAPTER             | CONTENT                                                                                    | FORMAT  | HOURS |
|-----|---------------------|--------------------------------------------------------------------------------------------|---------|-------|
| 7.1 | Proposal & Design   | Problem statement, choice of methods, design and evaluation plan, approved by the lecturer | Project | 3 h   |
| 7.2 | Development         | Implementation and experiments                                                             | Project | 3 h   |
| 7.3 | Evaluation & Report | Results, comparison with a baseline, written report                                        | Project | 1.5 h |
| 7.4 | Presentation        | Presentation to the class; questions and feedback                                          | Project | 1.5 h |

## FINAL PROJECT

## Capstone AI System

An AI system that combines at least two of the course's methods (search, game playing, logic, probabilistic or temporal reasoning) to solve a problem chosen by the team, evaluated against a baseline and explained in a written report.

**STACK** Python 3, Jupyter, NumPy, aima-python, NetworkX, pgmpy, hmmlearn

**DELIVERABLE** Proposal, code repository, written report, presentation